

Exercises 32.4-1 to 32.4-8

1 Exercise 32.4-1

Question 32.4-1

Compute the prefix function for the pattern

$$P = \text{ababbabbabbababbabb.}$$

Solution

Let

$$P = \text{ababbabbabbababbabb.}$$

Index the characters:

q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$P[q]$	a	b	a	b	b	a	b	b	a	b	b	a	b	a	b	b	a	b	b

The prefix-function values are:

$$\pi[1] = 0$$

because a single character has no proper prefix-suffix.

Proceeding through the pattern, we obtain:

q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\pi[q]$	0	0	1	2	0	1	2	0	1	2	0	1	2	3	4	5	6	7	8

So the prefix function is:

$$\pi = [0, 0, 1, 2, 0, 1, 2, 0, 1, 2, 0, 1, 2, 3, 4, 5, 6, 7, 8].$$

Final Answer

$$\pi = [0, 0, 1, 2, 0, 1, 2, 0, 1, 2, 0, 1, 2, 3, 4, 5, 6, 7, 8].$$

2 Exercise 32.4-2

Question 32.4-2

Give an upper bound on the size of $\pi^*[q]$ as a function of q . Give an example to show that your bound is tight.

Solution

The chain

$$\pi^*[q] = \{\pi[q], \pi[\pi[q]], \pi[\pi[\pi[q]]], \dots\},$$

strictly decreases, since

$$\pi[x] < x.$$

Starting below q , it can contain at most q values:

$$|\pi^*[q]| \leq q.$$

This is tight. For $P_q = \text{aaaaa}$ with $q = 5$,

$$\pi[5] = 4, \quad \pi[4] = 3, \quad \pi[3] = 2, \quad \pi[2] = 1, \quad \pi[1] = 0,$$

so

$$\pi^*[5] = \{4, 3, 2, 1, 0\},$$

whose size is $5 = q$.

Final Answer

$|\pi^*[q]| \leq q$, and the bound is tight for repeated-character patterns such as $\text{aaaa} \dots$.

3 Exercise 32.4-3

Question 32.4-3

Explain how to determine the occurrences of pattern P in the text T by examining the prefix function for the string PT (the concatenation of P and T).

Solution

Let $|P| = m$. Form $S = P\#T$, where $\#$ is a separator not in the alphabet, and compute the prefix function of S .

Whenever

$$\pi[i] = m,$$

the pattern P ends at position i of S . Since S starts with $P\#$, the corresponding shift in T is

$$i - 2m - 1$$

using 1-based indexing for S .

Final Answer

Compute π for $P\#T$. Every index i with $\pi[i] = |P|$ gives one occurrence of P in T .

4 Exercise 32.4-4

Question 32.4-4

Use an aggregate analysis to show that the running time of KMP-MATCHER is $\Theta(n)$.

Solution

The only suspicious part is the fallback loop:

```
while q > 0 and P[q+1] != T[i]
    q = pi[q]
```

During the whole scan:

- q increases at most n times, once per text character,
- each fallback strictly decreases q , since $\pi[q] < q$,
- q never becomes negative.

So the total number of fallbacks is also at most n . The outer loop has n iterations, hence the total work is $\Theta(n)$.

Final Answer

$\Theta(n)$, because q can increase at most n times and decrease at most n times.

5 Exercise 32.4-5

Question 32.4-5

Use a potential function to show that the running time of KMP-MATCHER is $\Theta(n)$.

Solution

Use the potential

$$\Phi = q,$$

where q is the number of currently matched pattern characters.

For one fallback $q \leftarrow \pi[q]$, the actual cost is 1, and the potential drops by at least 1. Thus its amortized cost is

$$1 + (\Phi_{\text{new}} - \Phi_{\text{old}}) = 1 + (\pi[q] - q) \leq 0.$$

For a successful match $q \leftarrow q + 1$, the amortized cost is at most

$$1 + 1 = 2.$$

There are at most n successful extensions and n outer-loop iterations, so the total amortized work is $O(n)$. Reading the text gives the lower bound $\Omega(n)$.

Final Answer

With potential $\Phi = q$, fallbacks pay for themselves by decreasing Φ . Hence KMP-MATCHER runs in $\Theta(n)$.

6 Exercise 32.4-6

Question 32.4-6

Show how to improve KMP-MATCHER by replacing the occurrence of π in line 7 (but not line 12) by π' , where π' is defined recursively for $q = 1, 2, \dots, m - 1$ by

$$\pi'[q] = \begin{cases} 0 & \text{if } \pi[q] = 0, \\ \pi'[\pi[q]] & \text{if } \pi[q] \neq 0 \text{ and } P[\pi[q] + 1] = P[q + 1], \\ \pi[q] & \text{if } \pi[q] \neq 0 \text{ and } P[\pi[q] + 1] \neq P[q + 1]. \end{cases}$$

Explain why the modified algorithm is correct, and in what sense it is an improvement.

Solution

The improved function π' skips fallback states that would fail on the same character. If

$$P[\pi[q] + 1] = P[q + 1],$$

then after a mismatch at state q , falling back to $\pi[q]$ would immediately mismatch again. So we jump farther:

$$\pi'[q] = \pi'[\pi[q]].$$

Otherwise the fallback state may be useful, so keep

$$\pi'[q] = \pi[q].$$

Replace only line 7 by:

```
q = pi'[q]
```

Keep line 12 unchanged:

```
q = pi[q]
```

π' preserves correctness because it skips only states that are guaranteed to fail immediately. Line 12 still uses π so overlapping matches are handled correctly.

Final Answer

Use π' only for mismatch fallback. The running time remains $\Theta(n)$, but the algorithm performs fewer redundant comparisons.

7 Exercise 32.4-7

Question 32.4-7

Give a linear-time algorithm to determine whether a text T is a cyclic rotation of another string T' . For example, **arc** and **car** are cyclic rotations of each other.

Solution

Two strings T and T' are cyclic rotations of each other if and only if:

- they have the same length,
- one appears as a substring of the other doubled.

1. If $|T| \neq |T'|$, return false.
2. Form $S = TT$.
3. Run KMP to test whether T' occurs in S .
4. If yes, return true; otherwise return false.

Example: $T = \text{arc}$, $TT = \text{arcarc}$, and $T' = \text{car}$ appears inside it. The running time is $O(n)$, because KMP searches a text of length $2n$.

Final Answer

Check whether T' occurs in TT using KMP. This solves the problem in $\boxed{O(n)}$ time.

8 Exercise 32.4-8

Question 32.4-8

Give an $O(m|\Sigma|)$ -time algorithm for computing the transition function δ for the string-matching automaton corresponding to a given pattern P . (Hint: Prove that $\delta(q, a) = \delta(\pi[q], a)$ if $q = m$ or $P[q + 1] \neq a$.)

Solution

First compute π in $\Theta(m)$. Then fill the transition table using:

$$\delta(q, a) = \begin{cases} q + 1, & \text{if } q < m \text{ and } P[q + 1] = a, \\ \delta(\pi[q], a), & \text{otherwise.} \end{cases}$$

COMPUTE-TRANSITION-FUNCTION-FAST(P , Σ)

```
1  m = |P|
2  compute pi[1..m]
3  for each a in Sigma
4      delta(0, a) = 0
5  delta(0, P[1]) = 1
6  for q = 1 to m
7      for each a in Sigma
8          if q < m and P[q+1] == a
9              delta(q, a) = q + 1
10         else
11             delta(q, a) = delta(pi[q], a)
12 return delta
```

If a extends the current match, move to $q + 1$. Otherwise, fall back to the largest border

state $\pi[q]$ and reuse its transition. There are $m + 1$ states and $|\Sigma|$ characters, so the table has $O(m|\Sigma|)$ entries.

Final Answer

Use the recurrence

$$\delta(q, a) = \begin{cases} q + 1, & \text{if } q < m \text{ and } P[q + 1] = a, \\ \delta(\pi[q], a), & \text{otherwise.} \end{cases}$$

After computing π , all entries are filled in $\boxed{O(m|\Sigma|)}$ time.